

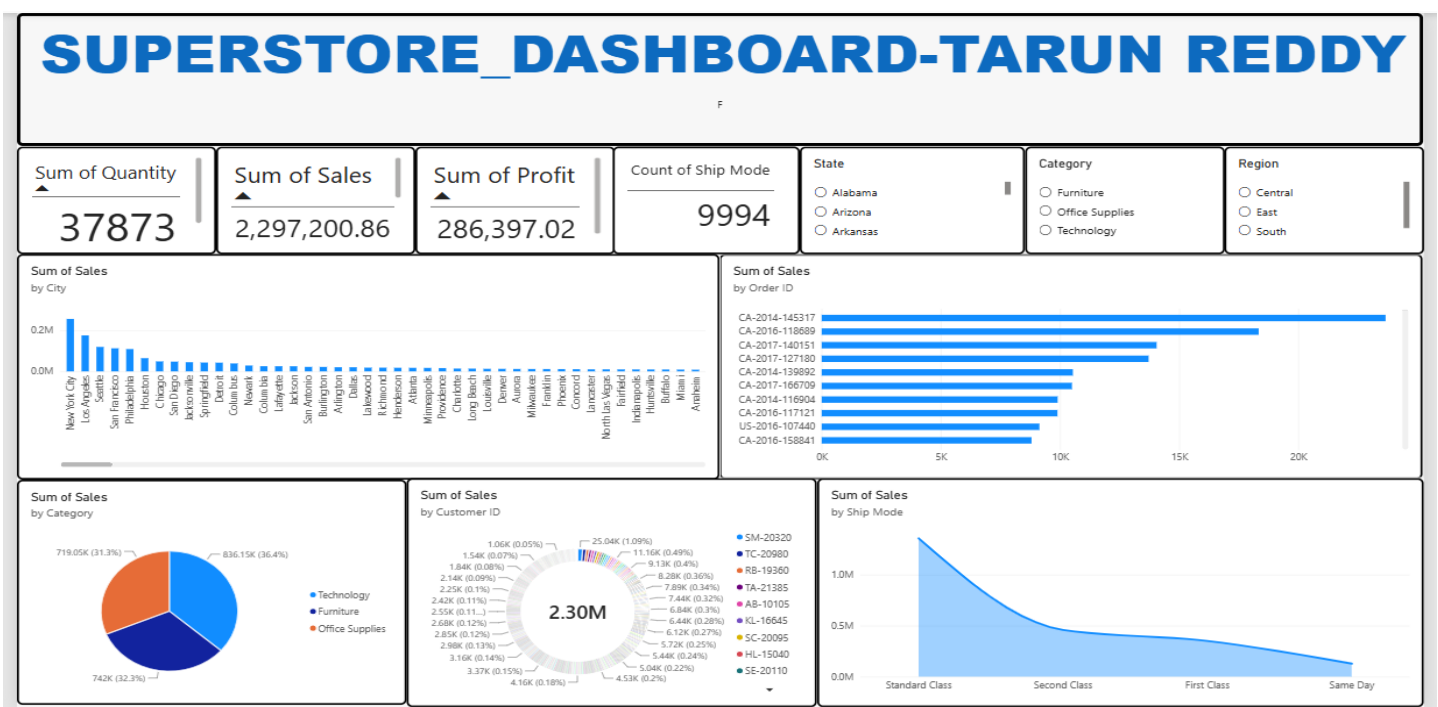
Apex Planet Data Analytics Internship —

Executive Summary

[V. Tarun Reddy] | Data Analytics Internship | [05/09/2026]

EXECUTIVE SUMMARY

This project analyzed the Superstore Sales dataset (9,994 orders, 21 features) to uncover key drivers of sales and profitability. Using Python, SQL, Power BI, and statistical modeling, the analysis identified that discounting is the single most consistent driver of reduced profit across products, sub-categories, and customer segments.



TOP 5 INSIGHTS

1. Technology is the most profitable category (\$145K), while Furniture lagssignificantly (\$18K) despite comparable sales volume.
2. Tables and Bookcases are net loss-making sub-categories (-\$17.7K and -\$3.5K), driven by discounts averaging 26% — more than double Technology's average discount.
3. West region leads in both sales and profit (\$725K sales), while South trails.
4. Customer segmentation revealed 4 distinct groups — a high-value VIP segment (avg.

profit \$2,229/customer) and a discount-sensitive segment that is unprofitable on average (avg. profit -\$95/customer).

5. Statistical testing confirmed a significant difference in profit between high- and low-discount orders (t-test, $p < 0.05$), and a significant relationship between Region and Category purchasing patterns (chi-square, $p < 0.05$).

BUSINESS RECOMMENDATIONS

1. Review discount policy for Tables, Bookcases, and the "Discount-Sensitive" customer segment — cap maximum discount thresholds or replace deep discounts with bundled offers to protect margin.
2. Prioritize retention programs for the VIP customer segment (~\$9,300 avg. spend, ~\$2,229 avg. profit) — this group drives disproportionate profitability.
3. Investigate regional strategy differences, particularly why the South region underperforms despite similar category mix — targeted marketing or logistics review may help close the gap.